

# Artificial intelligence is beginning to create value for selected small animal veterinary applications while remaining immature for others

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## Abstract

Artificial intelligence is a powerful technology with great potential to support veterinarians across many aspects of their multifaceted job. However, realizing this potential requires AI solutions that truly address specific pain points. Here we review 4 use cases in which AI has been applied to small animal veterinary medicine: image analysis, early disease detection, administration support, and disease surveillance. For each of these, we briefly present the current state of the technology and available AI applications. Results show that tangible value creation is heterogeneous across use cases and that availability of accurate and reliable data (eg, in the form of curated electronic health records) is a major limiting factor.

**Keywords:** artificial intelligence, diagnostic imaging, early disease detection, administrative workload, disease surveillance

## Introduction

Artificial intelligence is generally used to describe algorithms capable of producing output previously thought to require human intelligence.<sup>1</sup> It is a significant evolution from older machine learning algorithms that learn patterns from data in a human-assisted way.<sup>2,3</sup> Generative AI is a recent development, with algorithms that generate text or images from a simple request written in a natural language (the *prompt*).<sup>4</sup>

Technological innovations generally mature following a hype cycle<sup>5</sup> (**Figure 1**) characterized by an initial phase of inflated expectations (the *hype*), followed by a trough of disillusionment before tangible value creation takes off. Artificial intelligence is no exception. Backed by extensive research, AI has generated—and is still generating—massive expectations in many domains including human and veterinary medicine.<sup>6–12</sup> But where are we on tangible value creation in small animal veterinary medicine?

Value creation starts when technological innovations provide practical solutions to real-world problems or use cases. This review will assess current AI value creation in a diverse set of use cases. Findings

will be used to convey a realistic outlook and suggestions for strategies to speed up AI value creation in small animal veterinary medicine.

## Use Case 1: Diagnostic Imaging

Diagnostic imaging is one of the most common diagnostic tests in veterinary medicine. Unfortunately, the demand for veterinary radiologists outstrips their availability, leading to potential treatment delays and impacting patient welfare.<sup>13</sup> Teleradiology, in which the radiologist interprets images remotely, makes it easier to access radiologists but is not enough, as the low number of experts in many specialist areas still limits throughput. Therefore, obtaining rapid, high-quality specialist assessments of images remains a significant problem.

The computational discipline that designs AI algorithms for medical images is called *radiomics*.<sup>14</sup> Veterinary radiomics is already quite advanced, for two main reasons. Firstly, there are large amounts of accurate digitalized expert labeled data available for training new algorithms (ie, routinely collected radiographs that have a diagnosis label). Secondly, applications often build on highly optimized algorithms previously applied to other images or domains.<sup>15,16</sup> This has led to the publication of several high-performing algorithms that can identify potential quality problems with images<sup>17</sup> or can extract medically relevant information,<sup>18–24</sup> with several commercial solutions for image annotations now being available.<sup>25–29</sup>

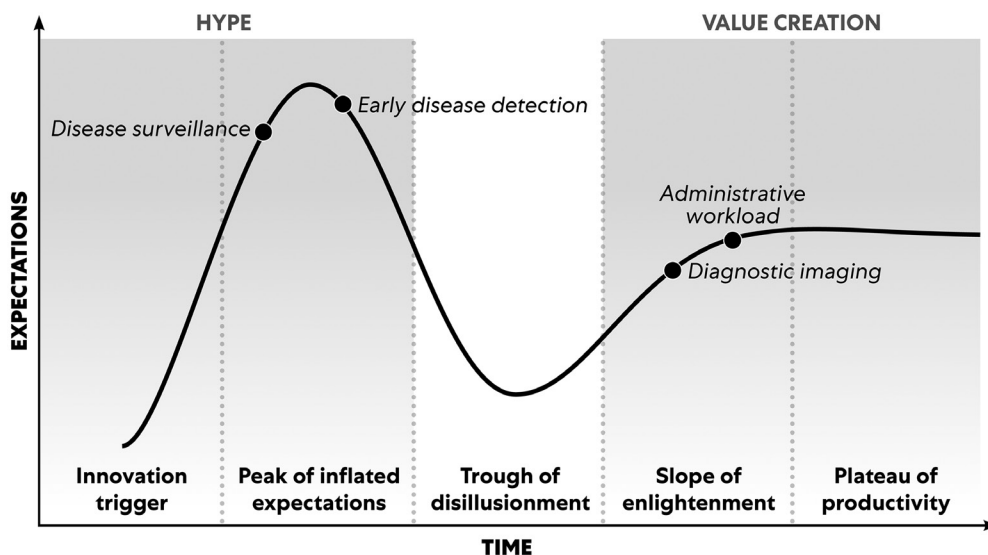
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**Figure 1**—Hype cycle for technological innovation<sup>5</sup> with the AI use cases discussed in this paper.

In terms of adoption, radiomics has the benefit that its outputs can be displayed on the image so that these can always be validated by a specialist whose feedback can also be used to improve the algorithms. This greatly alleviates common explainability concerns linked to the black box nature of AI models<sup>30–32</sup> and prevents being misguided by AI-induced artifacts or hallucinations. It is unlikely that radiomics will replace trained experts any time soon,<sup>33</sup> but it does already reduce their workload by simplifying the tasks of identifying important features in the images. Taken together, AI for diagnostic imaging is surely beyond the hype and is in the initial stages of adding tangible value to veterinary workflows.

## Use Case 2: Early Disease Detection

In current small animal veterinary medicine settings, many cases involve managing diseases already at advanced stages at the time of diagnosis. Earlier detection would obviously be desirable for potential improvements in prognosis, cost, and more timely, shorter, and less aggressive interventions. Early disease detection starts by individual-level risk models that allow veterinarians to identify pets at risk on the basis of demographics and clinical information so that, when appropriate, additional diagnostic tests can be performed during routine yearly checkups. Predictive models can then use all available information to support an early diagnosis. As it stands, these approaches, which are common in human medicine to predict mortality<sup>34,35</sup> or risk of cardiovascular diseases<sup>36</sup> or cancer,<sup>37</sup> for example, are rarely applied in small animal veterinary medicine.

Individual-level risk models estimate the risk of a specific individual to develop a medical condition at a given point in their life.<sup>38,39</sup> The development of accurate veterinary risk models requires data on pet signalments, health information, and other observations, with less common conditions generally requiring larger datasets.<sup>12,40,41</sup> Electronic medical record systems<sup>42–45</sup> are the most common source of such data,<sup>46</sup> with specific initiatives such as VetCompass<sup>47–49</sup> making such

datasets available to researchers. A few risk models have been developed for pets,<sup>47,48,50</sup> but important challenges remain. One of the main problems is that the data used can be biased due to being extracted from specific pet owner populations or due to unequal access to care, attitudes, economic status, or geographical location.<sup>51</sup> Therefore, advanced methodologies that are capable of explicitly addressing biases in the data should be considered to ensure the accuracy of models, such as Bayesian<sup>52–54</sup> or causal<sup>55,56</sup> approaches.

Early disease prediction models are types of individual-level risk models that use patient demographics supplemented with diagnostic test data, clinical signs, genetic data, or other pieces of information to support an early diagnosis of a disease with confidence and often do so before a reliable diagnosis can be obtained on the basis of clinical signs only. Artificial intelligence models have been developed to evaluate the risk of several veterinary conditions including anterior cruciate ligament rupture,<sup>57</sup> chronic kidney disease,<sup>58–61</sup> Cushing syndrome,<sup>62</sup> dermatitis,<sup>63</sup> epilepsy,<sup>64</sup> heart failure,<sup>65</sup> peritonitis,<sup>66</sup> hyperthyroidism,<sup>67</sup> and hypoadrenocorticism.<sup>68,69</sup> A major barrier to applying these in a clinical setting is the black box nature of many of these models, as they provide risk profiles without describing how or why a given result has been obtained.<sup>70–72</sup> Moreover, most of these models are only validated by the same researchers who created the model, often using data very similar to those used for model development. The lack of independent validation is a significant problem that affects veterinary and human models alike.<sup>73,74</sup> This concern is valid and real, as some human AI models have been found to be biased when applied in real-world scenarios.<sup>75,76</sup>

Overall, while progress is being made, the opportunities provided by AI have not yet moved the needle significantly for early disease detection in the clinic. Data availability is an important limiting factor that is apparent at the model-building stage, limiting performance, and at the independent model validation stage, limiting trust building. The black box nature of the AI models is an additional challenge that

limits veterinarians' understanding of the models on the journey to embrace them. Therefore, the use of AI in this space is mostly still hype. The scientific community is already addressing some of these concerns by developing technology to make the model explainable.<sup>77,78</sup> However, it will take some more time for this use case to mature, driven by gradual improvements in data collection and workflow adaptations that accommodate risk and early prediction models into clinical practice.

### Use Case 3: Administrative Workload

The burden of bureaucracy or tedious administration work have been described as major generators of veterinary workload,<sup>79</sup> with 40% of all veterinarians stating that administration was their number one problem.<sup>80</sup>

Artificial intelligence has the potential to automate many routine tasks such as scheduling appointments, managing patient records, taking consultation notes, tracking inventory, and handling basic customer inquiries. While developments are generally spearheaded in human medicine,<sup>6</sup> there is good progress in veterinary medicine-tailored solutions as well. For note-taking, the AI solutions are a combination of automatic speech recognition algorithms to transcribe conversation into text and natural language understanding algorithms to summarize key concepts of the text into clinical notes. Automatic speech recognition and natural language understanding are well established and widely used technologies that have allowed a speedy commercial deployment of a variety of tools.<sup>25,81-84</sup>

Artificial intelligence is also used to automate the patient-owner journey by scheduling and keeping track of appointments; automatically populating subjective, objective, assessment, and plan (SOAP) workflows; supporting veterinary decision-making; and sending invoices. This type of solution can be integrated into practice information management systems<sup>25,85-88</sup> or can be provided as plug-ins.<sup>89,90</sup> Some practice information management systems also have additional AI-based functionality for tracking inventory and measuring clinic performance.

In the context of this use case, the use of AI to communicate with pet owners is a particularly interesting topic. Artificial intelligence-based chatbots allow fast response at any time of the day, thus freeing up time from veterinarians and increasing pet owner engagement.<sup>91</sup> They can also be trained to be inclusive and avoid biases so that communication is more effective.<sup>92</sup> At the same time, there is a risk that chatbots provide inaccurate information and pet owners are uncomfortable with their messages being owned by a third party.<sup>93</sup>

Clearly, many AI-based solutions to perform administrative tasks are readily available to veterinarians. This type of AI is currently already creating value, and future work is likely to focus on refinement and improved IT and workflow integration.

### Use Case 4: Disease Surveillance

Disease surveillance, a cornerstone of veterinary epidemiology, aims to support the health of pet populations through the identification of outbreaks, monitoring of disease trends, and early detection of emerging threats. One of the main challenges is monitoring animal populations at a sufficiently large scale and high frequency given constraints on data collection, data access, and available resources. Several real-world data sources can be used to this end without the need for additional data collection,<sup>94,95</sup> such as primary care and referral clinical records,<sup>96</sup> pet insurance databases,<sup>97</sup> and internet search data.<sup>98</sup> However, the scale and complexity of these data make them unusable without the incorporation of AI.

In terms of supporting disease surveillance, AI can streamline data processing, knowledge extraction, quantification, and prediction.<sup>99-101</sup> A nice illustration is the work of the Small Animal Veterinary Surveillance Network<sup>101,102</sup> that used primary care clinical records to effectively track an outbreak of acute vomiting in dogs by means of unsupervised modeling and simple visualizations.<sup>103,104</sup> Artificial intelligence can also increase the usability of complex, nuanced free-text medical notes through processing via large language models specifically tailored for veterinary data such as DeepTag,<sup>105</sup> VetTag,<sup>106</sup> and PetBERT.<sup>95</sup> Confounding due to biases and missing data, which has the potential to limit reliable decision-making, can be addressed through causal inference methods, such as potential outcomes models, structural equation modeling, Bayesian methods, and directed acyclic graphs.<sup>107-112</sup>

While there are some good examples for where AI is starting to enable surveillance at scale, there remains scope and opportunity for it to create value. Analysis of real-world data presents as a huge opportunity; however, considerable effort and resources are still required to generate and manage the quality of these data.

### Conclusion and Outlook

Artificial intelligence value creation in small animal medicine is clearly not uniform across different veterinary applications, as highlighted by the use cases discussed (Figure 1). For diagnostic imaging and administrative workload, AI applications are mature and in the initial stages of value creation, as shown by commercial solutions being available as stand-alone services or as plug-ins that can be integrated into standard software. Early disease detection and disease surveillance are currently still in the hype phase, with lots of academic activity but limited real-world applications. As the underlying AI technology is quite similar, this poses the question of what drives the maturity differences.

In terms of value creation, it is important to understand that AI as a technology is merely a collection of models that become useful only after being trained on appropriate datasets. Consequently, use case-specific data are key to unlocking the value of AI, and that is where the heterogeneity clearly

emerges. For diagnostic imaging, abundant training data can be created with limited effort. Similarly, administrative-workload AI models either do not require specific training (eg, speech recognition models) or can be trained with available datasets (eg, clinical note vocabularies, SOAP manuals, diagnosis textbooks). The less mature early disease detection and disease surveillance use cases heavily depend on electronic health record data, and unfortunately that type of data is not available at a level of quantity and quality that is needed for effective real-world applications. Data challenges are linked to quantity (number of pets), quality (consistency of terminology), sparseness (big gaps due to limited number of visits), and coherence (unique pet identifiers across data sources). Low quantity and large sparseness will result in models unable to generalize properly to all expected scenarios, while low quality and low coherence will result in model outputs that may be inaccurate or even wrong. It thus seems that everything is ready for AI, except the data.

Leveraging AI in small animal medicine will require the profession to step up in terms of electronic health data collection and data sharing. While the technical approach to this has long been described,<sup>113</sup> who will drive the execution is less obvious. A collaborative consortium-based approach<sup>114</sup> like for human medicine is an option to consider and could involve academics, subject matter experts (eg, the Association for Veterinary Informatics), practicing veterinarians, pharmaceutical companies, kennel clubs, and pet owners. This will be a journey but one that can be a step change in the field.

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